Peru Inflation and Macroeconomic Stabilization (1980–2022).

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Introduction

This project analyzes inflation dynamics in Peru from 1980 to 2022 using publicly available World Bank data. Peru's macroeconomic history over this period is not characterized by a single stable inflation process. Instead, it includes a sharp hyperinflation episode in the late 1980s and early 1990s followed by a long period of low and relatively stable inflation after stabilization reforms.

The purpose of this project is not to estimate a causal model of inflation. Rather, it aims to answer a more practical analytical question:

How sensitive are simple inflation models to crisis years, regime shifts, and specification choices?

That framing matters because full-sample macro regressions can look statistically strong while being heavily driven by a few extreme observations. This report combines reproducible data collection, feature engineering, descriptive visualization, influence diagnostics, and specification comparisons to show why regime-aware analysis is necessary in small macroeconomic samples.

The dataset was built using the World Bank API through the `wbstats` R package. Annual observations were collected for Peru from 1980 to 2022.

Core indicators

Variable	Description
Inflation	CPI inflation (annual %)
Broad money growth	Broad money growth (annual %)
Exchange rate	Official exchange rate (LCU per USD)
Government consumption	Government final consumption expenditure (% GDP)
Terms of trade	Net barter terms of trade index

Engineered variables

In addition to the raw indicators, the analysis constructs:

- Exchange-rate depreciation: $100 \times \Delta \log(\text{exchange rate})$
- Terms-of-trade growth: $100 \times \Delta \log(\text{terms of trade})$
- Lagged macro variables
- Rolling inflation volatility
- Regime indicators, including a post-stabilization period and hyperinflation years
- Winsorized and transformed variants for robustness checks

Data Quality and Coverage

The table below documents the coverage of each World Bank series used in the analysis.

Table 2: Indicator coverage

Indicator	First year	Last year	Observations
Broad money growth (annual %)	1961	2022	62
Inflation, consumer prices (annual %)	1960	2024	65
General government final consumption expenditure (% of GDP)	1960	2024	65
Official exchange rate (LCU per US$, period average)	1960	2024	65
Net barter terms of trade index (2015 = 100)	1980	2023	44

Because annual growth rates and lagged variables are constructed from the raw series, some model specifications use slightly fewer observations than the full 1980–2022 span. Although some source series extend beyond 2022, the analytical sample is restricted to 1980–2022 for consistency across variables and specifications.

Analytical Approach

The project follows a descriptive, regime-aware workflow:

1. Build a reproducible macroeconomic panel from World Bank data.
2. Visualize the inflation series and identify the crisis regime.
3. Estimate a baseline full-sample model.
4. Audit whether a small number of years dominate the regression.
5. Compare coefficients across alternative samples and specifications.
6. Interpret results as descriptive associations rather than stable structural parameters.

This approach is more appropriate than relying on a single regression because Peru's inflation history includes an extreme structural break.

Inflation Regimes

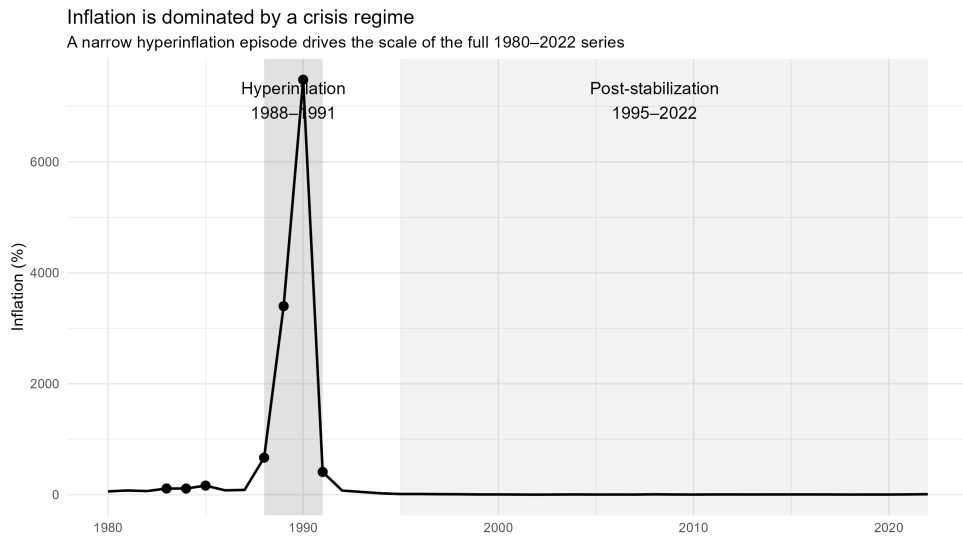


Figure 1: Inflation is dominated by a crisis regime

Figure 1 shows the central fact of the project: the full inflation series is dominated by a narrow hyperinflation episode between 1988 and 1991. The inflation spike around 1990 is so large that it determines the visual scale of the entire 1980–2022 sample.

This matters analytically because any full-sample regression is likely to be influenced heavily by these crisis years. It also suggests that Peru's inflation dynamics are better understood as a combination of at least two regimes:

- a crisis regime in the late 1980s and early 1990s
- a post-stabilization regime beginning in the mid-1990s

Summary Statistics

To show how strongly the crisis period affects the data, the table below reports summary statistics for the full sample and for a restricted sample excluding hyperinflation years.

Table 3: Panel A. Full sample summary statistics

Variable	N	Mean	SD	Min	Max
Inflation (CPI, %)	43	301.42	1237.50	0.19	7481.66
Broad money growth (%)	43	244.17	1005.03	-2.14	6384.92
FX depreciation (%)	42	39.05	86.51	-6.73	425.52
Terms-of-trade growth (%)	42	-0.64	10.23	-27.92	26.06
Government consumption (% GDP)	43	11.34	1.90	7.27	15.82

The contrast between the two panels is large. Full-sample moments are heavily distorted by the hyperinflation years, while the restricted sample provides a much better picture of typical macroeconomic variation.

Table 4: Panel B. Non-hyperinflation summary statistics

Variable	N	Mean	SD	Min	Max
Inflation (CPI, %)	36	17.21	26.78	0.19	85.82
Broad money growth (%)	36	26.63	28.51	-2.14	106.51
FX depreciation (%)	35	8.32	15.33	-6.73	50.18
Terms-of-trade growth (%)	35	0.20	10.47	-27.92	26.06
Government consumption (% GDP)	36	11.63	1.67	7.86	15.82

Baseline Model

The baseline specification is a simple descriptive model:

$$Inflation_t = \beta_0 + \beta_1 MoneyGrowth_t + \beta_2 FXDepreciation_t + \beta_3 ToTGrowth_t + \beta_4 GovConsumption_t + \epsilon_t$$

This equation is useful as a starting point, but it should not be interpreted as a structural model of inflation. In a small annual dataset with a major crisis episode, high in-sample fit can be misleading unless the model is checked for influence and stability.

Influence Diagnostics

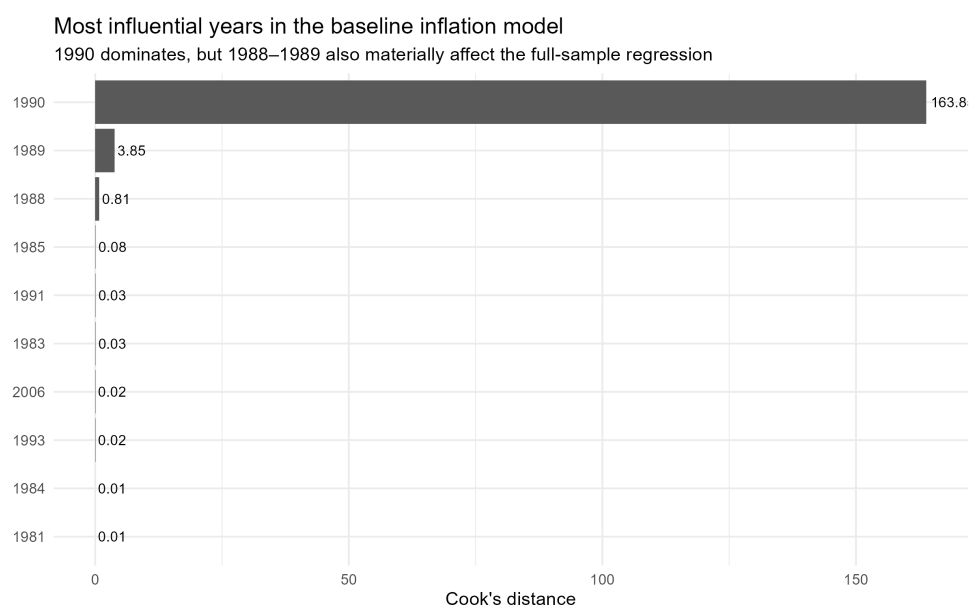


Figure 2: Most influential years in the baseline inflation model

Figure 2 shows that the baseline regression is dominated by a small number of years, especially 1990. That single observation is vastly more influential than the rest of the sample, while 1988 and 1989 also contribute meaningfully.

This result shows that the baseline full-sample regression is not a stable summary of the period. Rather than reflecting a relationship that holds uniformly across 1980–2022, the model is disproportionately shaped by a few crisis-era observations.

Top influential years

Table 5: Top influential years in the full-sample baseline model

year	inflation_cpi	broad_money_growth	fx_depr	.fitted	.resid	cooks_d	leverage	std_resid	dffits
1990	7481.664	6384.916	425.518	7651.776	-170.113	163.849	0.962	-5.686	-79.435
1989	3398.679	1917.282	302.990	2658.649	740.030	3.846	0.350	5.977	23.272
1988	667.019	620.951	203.500	1021.138	-354.118	0.813	0.337	-2.831	-2.248
1985	163.400	156.546	115.237	391.166	-227.766	0.083	0.139	-1.598	-0.657
1991	409.530	230.595	141.380	511.603	-102.072	0.031	0.216	-0.751	-0.392
1983	111.151	103.345	84.790	248.922	-137.771	0.029	0.134	-0.964	-0.379
2006	2.002	8.045	-0.664	-86.724	88.726	0.019	0.190	0.642	0.309
1993	48.580	71.797	46.748	150.244	-101.664	0.017	0.142	-0.715	-0.289
1984	110.209	127.219	75.551	247.163	-136.954	0.012	0.066	-0.923	-0.245
1981	75.433	68.678	37.983	153.212	-77.778	0.009	0.128	-0.542	-0.205

The table confirms that 1990 is an extreme leverage point in the regression. This supports the interpretation that naive full-sample OLS is fragile in this setting.

Specification Comparison

To assess whether the main relationships are stable, the project compares coefficients across four versions of the model:

- Full sample
- Full sample excluding hyperinflation years
- Pre-stabilization sample (1980–1994)
- Post-stabilization sample (1995–2022)

All comparisons below use Newey–West standard errors to make inference more appropriate for annual time-series data.

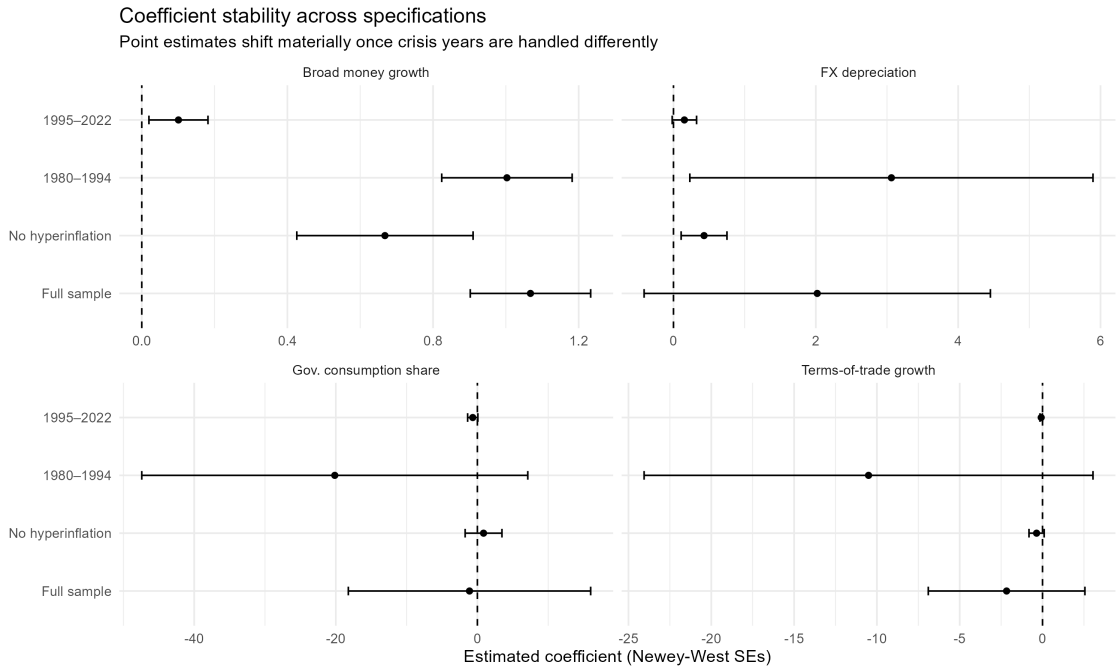


Figure 3: Coefficient stability across specifications

Figure 3 shows that coefficient magnitudes change materially across specifications.

Three patterns stand out:

1. Broad money growth remains positively associated with inflation in all four models, but the magnitude declines sharply in the post-stabilization period.
2. Exchange-rate depreciation follows a similar pattern: it is much more strongly associated with inflation in crisis-sensitive specifications than in the post-1995 sample.
3. Terms-of-trade growth and government consumption appear less stable and are estimated much less precisely.

This is the core analytical takeaway of the project. The sign pattern for money growth and exchange-rate depreciation is fairly persistent, but the magnitude of the relationship is regime-dependent.

Model Fit Statistics

Table 6: Model fit statistics across specifications

model	sample	outcome	n	r2	adj_r2	rmse	aic	bic
full_levels	full	inflation_cpi	42	0.986	0.985	144.136	548.734	559.160
no_hyper	inflation<100	inflation_cpi	35	0.883	0.868	8.812	263.657	272.989
pre_1995	1980-1994	inflation_cpi	14	0.989	0.984	210.872	201.565	205.400
post_1995	1995-2022	inflation_cpi	28	0.470	0.378	2.051	131.689	139.682
interaction	full	inflation_cpi	42	0.989	0.987	129.992	546.058	561.697
log_outcome	full_nonnegative	log1p_inflation	42	0.864	0.849	0.749	106.932	117.358
winsorized	full	inflation_cpi_w	42	0.981	0.979	23.629	396.839	407.265

The fit statistics reinforce the same point. R^2 remains very high in crisis-sensitive specifications, including the full-sample model, but falls materially in the post-stabilization sample. This suggests that the strong fit of the full-sample regression partly reflects the influence of crisis-era extremes rather than a stable inflation relationship across the entire 1980–2022 period.

The post-stabilization sample is especially informative: inflation is much lower and less volatile, and the same linear predictors explain a smaller share of variation. This suggests that the inflation process after stabilization differs meaningfully from the one observed during the crisis era.

Interaction Model

As an additional robustness check, the project estimates an interaction model that allows the slopes on money growth and exchange-rate depreciation to differ after stabilization.

Table 7: Interaction model with Newey–West standard errors

Term	Estimate	Std. Error	Statistic	p-value	CI low	CI high
(Intercept)	-162.946	133.023	-1.225	0.2290	-423.672	97.780
broad_money_growth	1.001	0.101	9.891	0.0000	0.802	1.199
post_stabilizationTRUE	212.941	104.029	2.047	0.0485	9.044	416.838
fx_depr	3.289	1.651	1.992	0.0544	0.053	6.526
tot_growth	-3.918	2.929	-1.338	0.1900	-9.659	1.822
gov_cons_gdp	-3.232	6.987	-0.463	0.6470	-16.926	10.462
broad_money_growth:post_stabilizationTRUE	-0.883	0.490	-1.802	0.0804	-1.843	0.077
post_stabilizationTRUE:fx_depr	-4.904	3.146	-1.559	0.1280	-11.071	1.262

The interaction model supports the same interpretation as the split-sample results: the association between inflation and broad money growth is much stronger before stabilization than after it. Exchange-rate effects also appear to weaken materially in the post-stabilization period, although these estimates are noisier and should be interpreted cautiously.

Main Findings

This project produces four main findings.

1. Peru's inflation history is not well described by a single stable regime

The 1988–1991 hyperinflation episode dominates the scale of the full sample and fundamentally changes how full-period regressions behave.

2. Full-sample OLS is highly sensitive to crisis years

Influence diagnostics show that the baseline model is driven disproportionately by 1990, with 1988 and 1989 also exerting meaningful leverage.

3. Broad money growth and exchange-rate depreciation are the most consistent correlates of inflation

These two variables remain positively associated with inflation across several specifications, but the estimated magnitudes shrink sharply once crisis years are excluded or the analysis is restricted to the post-stabilization period.

4. High fit does not imply stable inference

The full-sample model produces very high in-sample fit, but specification comparisons show that the apparent strength of the model is partly a consequence of crisis-era observations.

Limitations

This project is intentionally descriptive and has several limitations.

- The sample is small and annual, which limits statistical power.
- The pre-stabilization subsample is especially short.
- The analysis does not identify causal effects.
- Important omitted variables may still matter.
- Annual World Bank indicators cannot capture within-year crisis dynamics.

These limitations do not invalidate the project, but they do shape its interpretation. The contribution here is not causal identification. It is the careful auditing of model sensitivity in a historically unstable macroeconomic series.

Conclusion

This project began as a standard macroeconomic regression exercise, but the final analysis shows why that framing is too simplistic for Peru's inflation history.

A small number of crisis years dominate the full sample, strongly influence baseline regression estimates, and inflate the apparent explanatory power of naive OLS. Once those years are isolated, excluded, or modeled separately, coefficient magnitudes shift materially. Broad money growth and exchange-rate depreciation remain the strongest descriptive correlates of inflation, but their relationship with inflation is much weaker in the post-stabilization era than in the crisis regime.

This case study shows how a simple regression can become misleading in a regime-changing macro series.:

- build a reproducible data pipeline from public APIs
- engineer macroeconomic features
- identify structural breaks
- compare specifications rather than relying on a single model
- communicate model limitations clearly

That is ultimately the main analytical lesson of the project: in unstable macroeconomic data, the main lesson is that a high-fit regression can be misleading when a few crisis years dominate the sample.

Reproducibility

All data were downloaded from the World Bank API using R.
All figures, tables, and model outputs were generated through a reproducible script-based workflow.

The full repository includes:

- data download and cleaning
- feature engineering
- model estimation

- robustness checks
- figure and table generation

Appendix

Appendix A. Macroeconomic dashboard

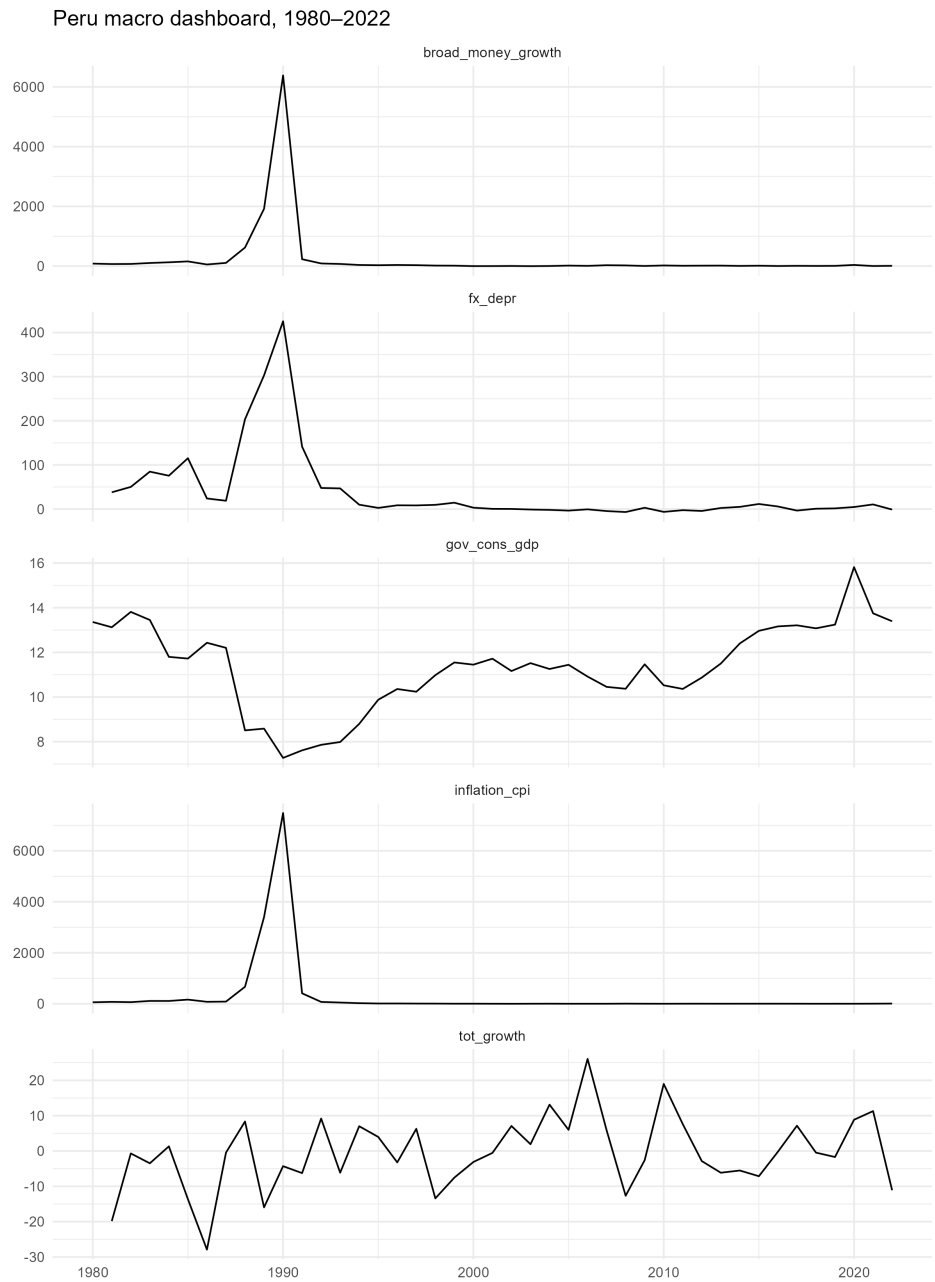
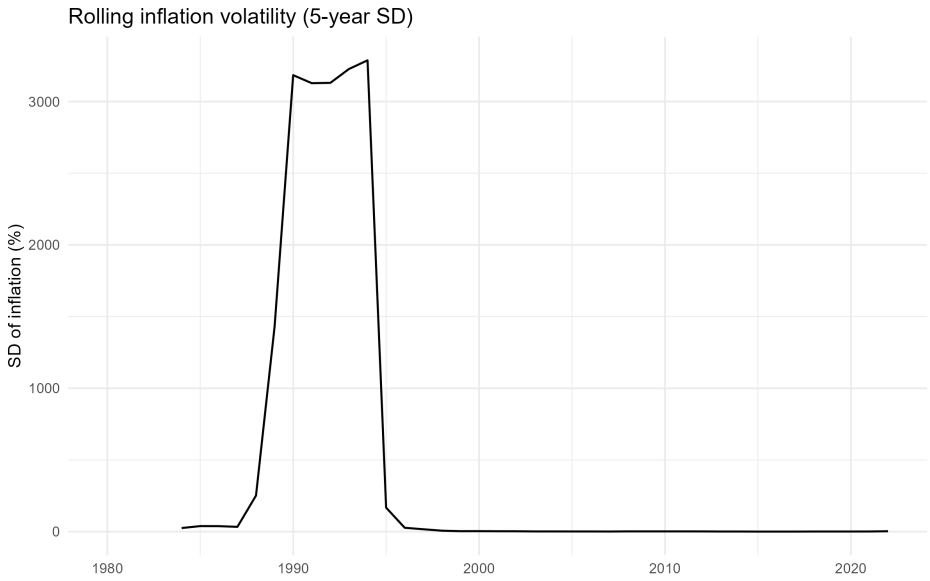


Figure 4: Macroeconomic dashboard for Peru, 1980–2022

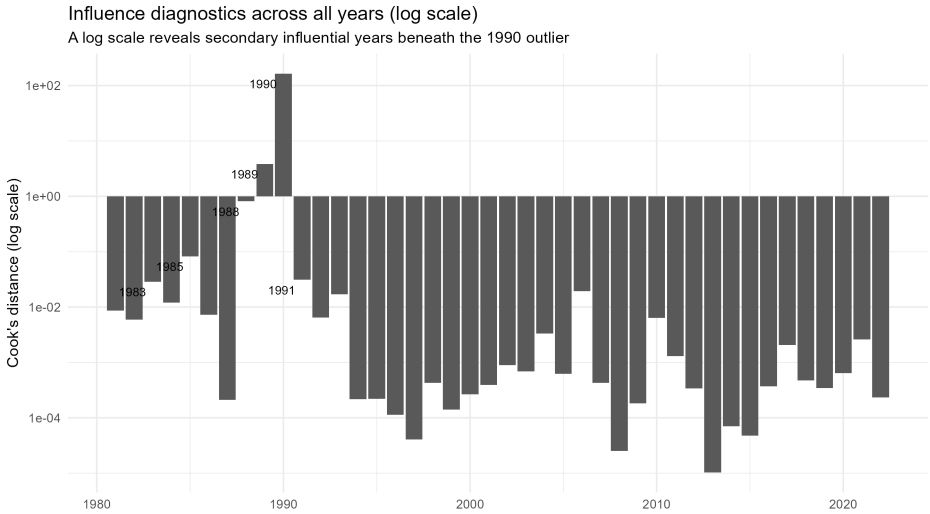
This dashboard provides broader macroeconomic context for the period under study. It is useful descriptively, but the main report emphasizes the inflation regime chart because it communicates the structural-break problem more directly.

Appendix B. Rolling inflation volatility



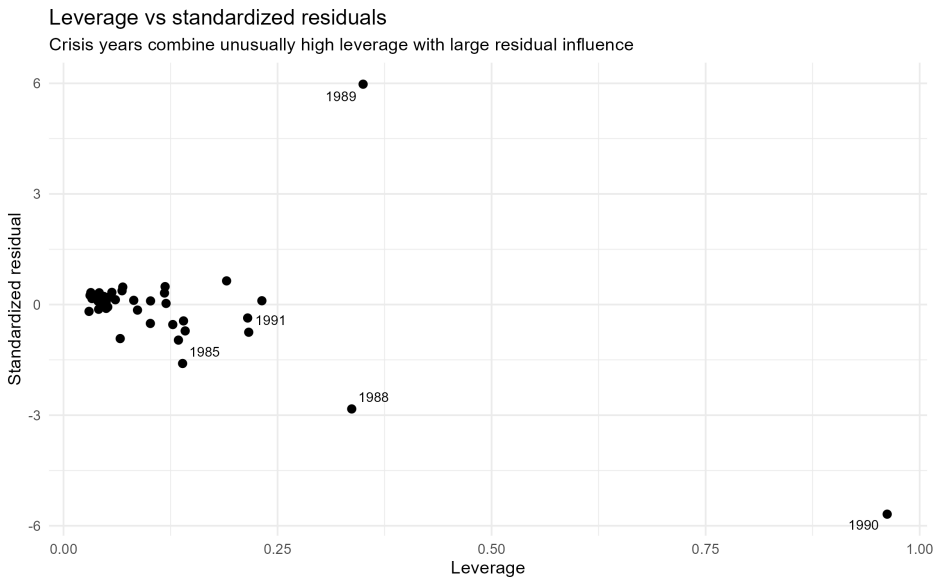
This figure complements the regime chart by showing how inflation volatility collapses after stabilization. It reinforces the interpretation that Peru's macroeconomic environment became much more stable after the crisis era.

Appendix C. Full-sample influence diagnostics (log scale)



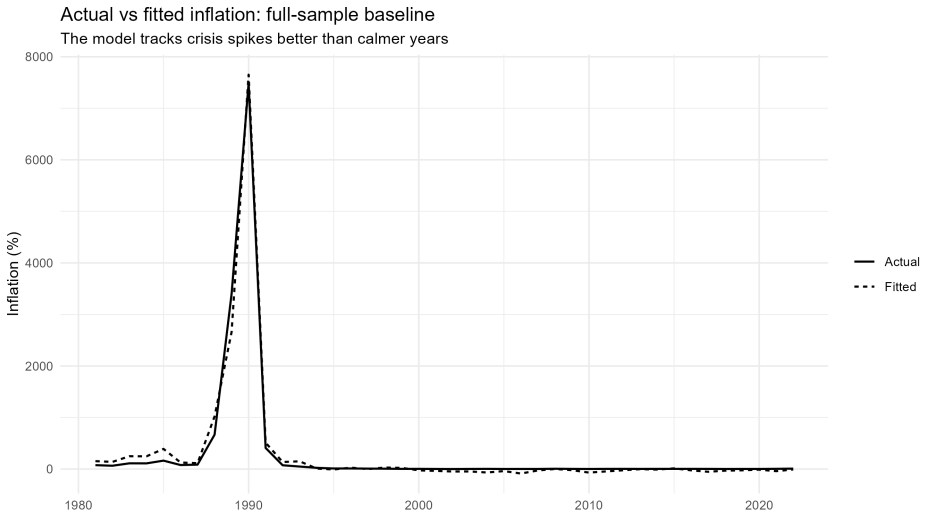
The main report uses the top-10 influential-years chart for readability. This log-scale figure provides a fuller diagnostic view of the influence distribution across all years.

Appendix D. Leverage and standardized residuals



This diagnostic plot shows that crisis years combine unusually high leverage with large residual influence, reinforcing the conclusion that the baseline full-sample regression is not a stable summary of the period.

Appendix E. Actual versus fitted inflation



This figure shows that the baseline model tracks crisis spikes much more closely than calmer post-stabilization years. It is therefore better read as a descriptive fit to crisis-era variation than as evidence of a stable inflation equation.